

$$1) \text{ On a } \int_0^{+\infty} e^{-at} \frac{\sin bt}{t} dt$$

$$= \int_0^1 e^{-at} \frac{\sin bt}{t} dt + \int_1^{+\infty} e^{-at} \frac{\sin bt}{t} dt$$

$$\text{On a } \lim_{t \rightarrow 0} e^{-at} \frac{\sin bt}{t} = \lim_{t \rightarrow 0} b e^{-at} \frac{\sin bt}{bt}$$

$$= b \text{ car } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{Donc } \int_0^1 e^{-at} \frac{\sin bt}{t} dt \text{ converge } (1)$$

$$\text{Pour } \int_1^{+\infty} e^{-at} \frac{\sin bt}{t} dt$$

$$\text{On a } |\sin bt| \leq 1$$

$$\left| e^{-at} \frac{\sin bt}{t} \right| \leq \frac{e^{-at}}{t} \leq e^{-at}$$

$$\int_1^{+\infty} \left| e^{-at} \frac{\sin bt}{t} \right| dt \leq \int_1^{+\infty} e^{-at} dt$$

$$\text{On a } \int_1^{+\infty} e^{-at} dt = \lim_{x \rightarrow +\infty} \int_1^x e^{-at} dt$$

$$\Rightarrow \int_1^{+\infty} e^{-at} dt = \lim_{x \rightarrow +\infty} \left[-\frac{e^{-at}}{a} \right]_1^x$$

$$\Rightarrow \int_1^{+\infty} e^{-at} dt = \lim_{x \rightarrow +\infty} \left(-\frac{e^{-ax}}{a} + \frac{e^{-at}}{a} \right)$$

$$\Rightarrow \int_1^{+\infty} e^{-at} dt = \frac{e^{-a}}{a} \in \mathbb{R}$$

$$\text{Donc } \int_1^{+\infty} e^{-at} dt \text{ converge}$$

$$\text{Donc } \int_1^{+\infty} \left| e^{-at} \frac{\sin bt}{t} \right| dt$$

absolument converge

$$\Rightarrow \int_1^{+\infty} e^{-at} \frac{\sin bt}{t} dt \text{ converge } (2)$$

D'après (1) et (2) on conclut

que $\int_0^{+\infty} e^{-at} \frac{\sin bt}{t} dt$ est convergent

2-a) D'après 1) On a

$$\int_0^{+\infty} e^{-at} \frac{\sin bt}{t} dt \text{ est bien}$$

défini sur $\mathbb{R}_+^*$

$$b). \text{ Soit } F(x) = \int_0^{+\infty} e^{-at} \frac{\sin bt}{t} dt$$

$$\text{Posons } f(x, t) = e^{-at} \frac{\sin xt}{t}$$

$$\text{On a } \lim_{x \rightarrow +\infty} f(x, t) = 0 \text{ car}$$

$$0 \leq \left| e^{-at} \frac{\sin bt}{t} \right| \leq \frac{|e^{-at}| |\sin bt|}{|t|}$$

$$\leq \frac{e^{-at}}{t} \cdot 1 = \frac{e^{-at}}{t}$$

En particulier quand $t \rightarrow +\infty$

$$\text{D'où } \lim_{x \rightarrow +\infty} f(x, t) = 0$$

$$\Rightarrow \int_0^{+\infty} e^{-at} \frac{\sin xt}{t} dt \xrightarrow{x \rightarrow +\infty} 0$$

$$\Rightarrow \lim_{x \rightarrow +\infty} F(x) = 0$$

Calculons $F'(x)$

$$F'(x) = \int_0^{+\infty} \frac{\partial f(x,t)}{\partial x} dx$$

$$\Rightarrow F'(x) = \int_0^{+\infty} \cos(xt) e^{-at} dt = \int_0^{+\infty} g(x,t) dt$$

De même $\lim_{x \rightarrow +\infty} g(x,t)$

$$= \lim_{x \rightarrow +\infty} \cos(xt) e^{-at} = 0$$

car $\forall t > 0 ; 0 < |\cos(xt) e^{-at}| < e^{-at}$

$$\text{Donc } \lim_{x \rightarrow +\infty} |\cos(xt) e^{-at}| = 0$$

$$\text{car } \lim_{t \rightarrow +\infty} e^{-at} = 0$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \cos(xt) e^{-at} = 0$$

$$\Rightarrow \int_0^{+\infty} \cos(xt) e^{-at} dt \xrightarrow{x \rightarrow +\infty} 0$$

$$\Rightarrow \lim_{x \rightarrow +\infty} F'(x) = 0$$

$$3-a) \text{ On a } F'(x) = \int_0^{+\infty} e^{-at} \cos(xt) dt$$

$$\Rightarrow F'(x) = \int_0^{+\infty} e^{-at} \operatorname{Re}(e^{ixt}) dt$$

$$\Rightarrow F'(x) = \int_0^{+\infty} \operatorname{Re}(e^{-at} e^{ixt}) dt$$

$$\Rightarrow F'(x) = \operatorname{Re} \left(\int_0^{+\infty} e^{-at} e^{ixt} dt \right)$$

$$\Rightarrow F'(x) = \operatorname{Re} \left(\int_0^{+\infty} e^{-(a-ix)t} dt \right)$$

$$\text{On a } \int_0^{+\infty} e^{-(a-ix)t} dt = \lim_{c \rightarrow +\infty} \int_0^c e^{-t(a-ix)} dt$$

$$= \lim_{c \rightarrow +\infty} \left[\frac{-1}{a-ix} e^{-t(a-ix)} \right]_0^c$$

$$= \lim_{c \rightarrow +\infty} \frac{-1}{a-ix} e^{-c(a-ix)} + \frac{1}{a-ix}$$

$$\text{or } \left| \frac{-1}{a-ix} \right| \cdot x |e^{-ca}| \cdot x |e^{icx}| = \left| \frac{-1}{a-ix} \right| \cdot e^{-ca}$$

$$\text{Donc } \lim_{c \rightarrow +\infty} \frac{-1}{a-ix} e^{-c(a-ix)} + \frac{1}{a-ix}$$

$$= \frac{1}{a-ix} = \frac{1 \times (a+ix)}{(a-ix)(a+ix)} = \frac{a+ix}{a^2 - (ix)^2}$$

$$= \frac{a+ix}{a^2 + x^2} = \frac{a}{a^2 + x^2} + i \frac{x}{a^2 + x^2}$$

$$\text{Donc } F'(x) = \operatorname{Re} \int_0^{+\infty} e^{-at} \cos(xt) dt = \frac{a}{a^2 + x^2}$$

$$\text{D'où } F'(x) = \frac{a}{a^2 + x^2}$$

$$\Rightarrow \int F'(x) = \int \frac{a}{a^2 + x^2}$$

$$\Rightarrow F(x) = \int \frac{1}{a(1 + (\frac{x}{a})^2)}$$

$$\Rightarrow F(x) = \int \frac{1/a}{1 + (\frac{x}{a})^2} = \int \frac{(\frac{x}{a})'}{1 + (\frac{x}{a})^2}$$

$$\Rightarrow F(x) = \operatorname{Arctan} \left(\frac{x}{a} \right) + c$$

$$\text{on sait } \left\{ \begin{array}{l} \lim_{x \rightarrow +\infty} F(x) = 0 \\ \lim_{x \rightarrow +\infty} \operatorname{Arctan} \frac{x}{a} = \frac{\pi}{2} \end{array} \right.$$

$$\Rightarrow 0 = \frac{\pi}{2} + c \Rightarrow c = -\frac{\pi}{2}$$

$$\Rightarrow 0 = \frac{\pi}{2} + c \Rightarrow c = -\frac{\pi}{2}$$

$$\text{Donc } F(x) = \operatorname{Arctan} \left(\frac{x}{a} \right) - \frac{\pi}{2}$$

$$3-b). \text{ On a } F(x) = \text{Arctan}\left(\frac{x}{a}\right) - \frac{\pi}{2}$$

$$\Rightarrow F(a) = \text{Arctan } 1 - \frac{\pi}{2}$$

$$\Rightarrow F(a) = \frac{\pi}{4} - \frac{\pi}{2}$$

$$\Rightarrow F(a) = -\frac{\pi}{4}$$

$$\begin{aligned} & \cdot \text{ On a } \frac{\pi}{2a} (x^2 + a^2) F'(x) + F(x) \\ &= \frac{\pi}{2a} (x^2 + a^2) \cdot \frac{a}{(x^2 + a^2)} + \text{Arctan} \frac{x}{a} - \frac{\pi}{2} \\ &= \frac{\pi}{2} + \text{Arctan} \frac{x}{a} - \frac{\pi}{2} \\ &= \text{Arctan} \frac{x}{a} \end{aligned}$$

$$4) \text{ On a } \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt$$

d'après 1) cet integrale existe

$$\begin{aligned} & \text{On pose } T = \sqrt{2}t \\ & \Rightarrow dT = \sqrt{2}dt \end{aligned}$$

$$= \int_0^{+\infty} e^{-T} \frac{\sin T}{T} dT$$

Donc

$$\int_0^{+\infty} e^{-t} \frac{\sin t}{t} dt = \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt$$

$$\text{On a } \int_0^{+\infty} e^{-at} \frac{\sin xt}{t} dt = F(x) = \text{Arctan} \frac{x}{a} - \frac{\pi}{2}$$

$$\text{pour } a = x = \sqrt{2}$$

$$\int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt = F(\sqrt{2})$$

$$\Rightarrow \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt = \text{Arctan} \frac{\sqrt{2}}{\sqrt{2}} - \frac{\pi}{2}$$

$$\Rightarrow \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt = \text{Arctan } 1 - \frac{\pi}{2}$$

$$\Rightarrow \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt = \frac{\pi}{4} - \frac{\pi}{2}$$

$$\Rightarrow \int_0^{+\infty} e^{-\sqrt{2}t} \frac{\sin \sqrt{2}t}{t} dt = -\frac{\pi}{4}$$

$$\cdot \text{ Pour } a = x = 1$$

$$\int_0^{+\infty} e^{-t} \frac{\sin t}{t} dt = F(1) = \text{Arctan} \frac{1}{1} - \frac{\pi}{2}$$

$$\Rightarrow \int_0^{+\infty} e^{-t} \frac{\sin t}{t} dt = -\frac{\pi}{4}$$

$$5) \frac{(x^2 + a^2)}{a} y'(x) + \frac{2}{\pi} y(x) = \frac{2}{\pi} \text{Arctan} \frac{x}{a}$$

$$\text{On divise par } \frac{x^2 + a^2}{a}$$

$$\Rightarrow y'(x) + \frac{2}{\pi} \cdot \frac{a}{x^2 + a^2} y(x) = \frac{2}{\pi} \cdot \frac{a}{x^2 + a^2} \text{Arctan} \frac{x}{a}$$

La solution de l'equation homogene

$$y' + \frac{2}{\pi} \frac{a}{x^2 + a^2} y = 0$$

est donnee par $y_H = e^{-\int \frac{2a}{\pi(x^2 + a^2)} dx}$

et $A(x)$ est une primitive de $\frac{2}{\pi} \cdot \frac{a}{x^2 + a^2}$

$$A(x) = \int \frac{2}{\pi} \cdot \frac{a}{x^2 + a^2} dx$$

$$\Rightarrow A(x) = \frac{2}{\pi} \int \frac{a}{a^2 \left(\left(\frac{x}{a} \right)^2 + 1 \right)} dx$$

$$\Rightarrow A(x) = \frac{2}{\pi} \int \frac{1/a}{1 + \left(\frac{x}{a} \right)^2}$$

$$\Rightarrow A(x) = \frac{2}{\pi} \text{Arctan} \frac{x}{a}$$

$$\text{Donc } y_H = K e^{-\frac{2}{\pi} \operatorname{Arctan} \frac{x}{a}}$$

cherchons y_P

$$E: \frac{x^2 + a^2}{a^2} y' + \frac{2}{\pi} y = \frac{2}{\pi} \operatorname{Arctan} x$$

$$\Rightarrow \frac{2}{\pi} \frac{x^2 + a^2}{a^2} y' + y = \operatorname{Arctan} x$$

D'après 3-b) On a :

$$\frac{\pi}{2} \frac{x^2 + a^2}{a} F'(x) + F(x) = \operatorname{Arctan} x$$

$\Rightarrow F$ est une solution particulière

D'où :

$$y = K e^{-\frac{2}{\pi} \operatorname{Arctan}(\frac{x}{a})} + F(x)$$

